

SIH 2026 — ISBM College of Engineering — Internal Hackathon

Statement_ID	Category	Theme	Problem Statement Title	Description
1	Software	Agriculture, FoodTech & Rural Development	Post-Harvest Loss Prediction and Cold-Chain Gap Mapping Platform	A significant share of India's fresh produce is lost between farm and market due to fragmented cold-chain infrastructure. Design a platform that maps existing cold-storage capacity against real harvest volumes, predicts where loss risk is highest, and recommends mobile/pop-up cold-storage deployment and route optimization for perishable transport.
2	Software	Agriculture, FoodTech & Rural Development	Groundwater Depletion Early-Warning System Using Crowd-Sourced Well Data	Groundwater monitoring is sparse in many rural blocks, and depletion often becomes visible only after wells run dry. Build a system where hand-pump/well readings crowd-sourced by local volunteers combine with satellite soil-moisture data to flag blocks approaching critical depletion months ahead.
3	Software	MedTech / BioTech / HealthTech	Early Sepsis Risk Detection for Rural PHCs Using Vital-Sign Trends	Sepsis in Primary Health Centre settings is frequently caught late due to lack of continuous monitoring. Build a lightweight system that tracks basic vitals over time, available even without ICU-grade equipment, and flags deteriorating trends before a patient needs referral.
4	Software	MedTech / BioTech / HealthTech	Postpartum Mental Health Screening and Referral Pathway for ASHA Workers	Postpartum depression in rural India is under-screened due to stigma and lack of structured tools for frontline workers. Build a stigma-sensitive digital screening flow that ASHA/ANM workers can use during routine visits, with a clear referral pathway to available care.
5	Software	Smart Education	Regional-Language Doubt-Solving Assistant for STEM Without Internet Dependency	Many doubt-solving apps require constant connectivity and are English-first, excluding students in low-connectivity regional-language contexts. Build an offline-capable or low-bandwidth assistant that explains STEM concepts step-by-step in regional languages.
6	Hardware	Smart Education	Braille-Integrated Low-Cost Tablet for Visually Impaired Students in Government Schools	Assistive learning devices for visually impaired students remain too expensive for most government schools to procure at scale. Design a low-cost device combining refreshable Braille output with audio, usable for basic literacy and STEM learning.
7	Software	Disaster Management & Climate Resilience	Post-Disaster Structural Damage Triage Using Crowd-Sourced Images	After earthquakes or cyclones, structural safety assessment of thousands of buildings is slow and resource-constrained. Build a system that lets citizens upload building photos, using image analysis to give a rapid first-pass safety triage to prioritize official inspection.
8	Software	Disaster Management & Climate Resilience	Heatwave Vulnerability Mapping for Outdoor and Gig Workers	Heatwave advisories are generic and don't account for outdoor/gig workers' actual exposure patterns. Build a system that maps real-time heat-risk against opt-in worker location/schedule data and pushes targeted rest/hydration alerts to the most exposed workers.
9	Software	Blockchain & Cybersecurity	Deepfake Detection Tool Tailored to Regional-Language Scams	Deepfake detection tools are largely trained on English/Western datasets and miss regional-language voice-cloning scams common in India. Build a detection tool specifically trained to catch manipulated audio/video in Indian regional languages and accents.
10	Software	Blockchain & Cybersecurity	Dark-Web Marketplace Monitoring Tool for Counterfeit Medicine Sales	Counterfeit medicines increasingly move through encrypted/dark-web channels current monitoring tools aren't built to track. Build a monitoring and alerting tool, within legal and ethical bounds, that flags patterns indicative of counterfeit pharma sales for law-enforcement follow-up.

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11	Software	Transportation & Logistics	Multi-Modal Public Transit Planner for Tier-2/3 Cities Without Unified APIs	Unlike metro cities, Tier-2/3 cities often lack a single unified transit data source, making multi-modal trip planning impossible for citizens. Build a planner that aggregates fragmented, informal transit data into a usable trip-planning tool.
12	Hardware	Transportation & Logistics	Solar-Assisted Cold-Box for Two-Wheeler-Based Last-Mile Perishable Delivery	Two-wheeler delivery of perishables such as dairy and medicines currently has no reliable temperature control. Design a compact, solar-assisted cold-box attachable to delivery two-wheelers that maintains cold-chain integrity for short-duration trips.
13	Software	Smart Automation	Voice-Command Interface for Assembly-Line Workers with Limited Digital Literacy	Digital shop-floor tools often require typing/touchscreen interaction that slows down or excludes workers with limited digital literacy. Build a voice-command interface in regional languages for logging production status, defects, and requests.
14	Software	Smart Automation	Automated Compliance-Gap Checker for MSMEs Against Evolving Labour and Safety Regulations	MSMEs frequently fall out of compliance simply because regulations change faster than they can track. Build a tool that cross-references a unit's declared processes against current regulations and flags specific compliance gaps.
15	Software	Renewable / Sustainable Energy	E-Waste Collection Route Optimization Platform Connecting Households to Certified Recyclers	Formal e-waste recycling remains under-accessed because pickup logistics from households to certified recyclers are fragmented. Build a platform that aggregates household e-waste pickup requests and optimizes routes to certified recycling centers.
16	Software	Travel & Tourism / Heritage & Culture	Overtourism Load Balancing System for Fragile Heritage and Ecological Sites	Popular heritage and ecological sites face degradation from unmanaged visitor overcrowding during peak periods. Build a system that predicts crowding and dynamically recommends alternate timing, routes, or nearby lesser-known sites.
17	Hardware	Travel & Tourism / Heritage & Culture	Low-Cost Structural Health Monitoring Sensors for Ageing Heritage Structures	Many heritage structures lack any structural monitoring, and deterioration is caught only after visible damage. Design a low-cost, non-invasive sensor kit to monitor vibration, tilt, and moisture trends without altering the structure's fabric.
18	Hardware	Travel & Tourism / Heritage & Culture	Solar-Powered Smart Signage and Wayfinding Kiosks for Rural Tourism Circuits	Rural tourism circuits often lack reliable signage and wayfinding infrastructure, and grid power isn't always available at remote points. Design solar-powered, low-maintenance smart signage kiosks giving directions and basic site information.
19	Software	Water & Sanitation	Real-Time Water Quality Contamination Source-Tracing for Municipal Supply Networks	When contamination is detected in a municipal water network, tracing its source through the distribution system is slow and manual. Build a system using sensor placement and flow modeling to narrow down likely contamination-source segments quickly.
20	Software	Governance & Public Services	Grievance Redressal Bottleneck Analyzer for State-Level Public Service Portals	State grievance portals accumulate large backlogs, but there's little visibility into where in the process bottlenecks actually occur. Build an analytics tool identifying which stages or departments cause the most delay so resources can be targeted there.

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21	Hardware	Governance & Public Services	Solar-Powered Digital Notice Board for Panchayat-Level Information Dissemination	Panchayat-level information on schemes, meetings, and alerts is often disseminated inconsistently through physical notices that get missed. Design a low-cost, solar-powered, remotely-updatable digital notice board for village centers.
22	Software	Robotics and Drones	Drone Flight-Path Planning Tool for Search-and-Rescue in Dense Forest Canopy	Search-and-rescue drones struggle with dense forest canopy blocking both visual search and GPS signal. Build a flight-path planning and signal-relay coordination tool optimized for canopy-obstructed search operations.
23	Hardware	Robotics and Drones	Low-Cost Amphibious Rescue Robot for Flood-Affected Areas	Flood rescue in areas combining debris, shallow water, and mud is dangerous for human responders and poorly served by existing rescue robots. Design a low-cost amphibious robot capable of navigating mixed flood terrain to locate stranded individuals.
24	Software	Governance & Public Services	Public Infrastructure Project Delay Prediction Using Historical Tendering and Execution Data	Public infrastructure projects such as roads and schools often overrun timelines with little early warning. Build a prediction model using historical tendering, contractor, and execution-pace data to flag high delay-risk projects early.
25	Software	Robotics and Drones	Swarm Coordination Software for Multi-Drone Agricultural Spraying Over Irregular Plots	Multi-drone spraying coordination software assumes regular field shapes, which doesn't match many Indian smallholder plot geometries. Build swarm-coordination software that adapts spray paths to irregular, small, and adjacent-plot boundaries.
26	Software	Financial Inclusion & FinTech	Savings-Nudge App for Informal-Sector Workers with Irregular Income	Standard savings apps assume regular salaried income, which doesn't fit informal-sector workers with irregular cash flow. Build a savings tool that adapts nudges and targets to genuinely irregular income patterns rather than fixed monthly assumptions.
27	Software	Financial Inclusion & FinTech	Fraud Detection for Micro-Loan Groups Using Group Behaviour Patterns	Self-help-group and joint-liability-group micro-lending is vulnerable to group-level fraud patterns such as ghost members and cross-guaranteeing that individual-level fraud tools miss. Build a detection tool specifically modeling group-lending behaviour.
28	Software	Fitness & Sports	Injury-Risk Prediction Tool for School-Level Athletes Using Simple Movement Data	School-level sports programs rarely have access to injury-prevention tools that professional teams use. Build a lightweight injury-risk prediction tool using simple camera- or wearable-based movement data, tuned for school-budget contexts.
29	Software	Space Technology & Emerging Tech	Space-Debris Collision-Risk Visualization Dashboard for Small Satellite Operators	Small satellite operators from academia and startups often lack accessible tools to assess collision risk from tracked debris. Build a simplified dashboard translating public debris-tracking data into actionable collision-risk alerts.
30	Software	Space Technology & Emerging Tech	Synthetic-Media Literacy Training Tool to Help Citizens Identify AI-Generated Misinformation	As AI-generated text, image, and video misinformation grows, most citizens have no structured way to build detection literacy. Build an interactive training tool teaching practical techniques to spot common signs of synthetic or manipulated media.

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31	Software	Agriculture, FoodTech & Rural Development	AI-Based Crop Disease Early Warning System for Marginal Farmers	A mobile app that lets farmers upload leaf photos for instant disease diagnosis, cross-referenced with local weather data to predict outbreak risk in the surrounding area.
32	Software	MedTech / BioTech / HealthTech	Unified Digital Health Passport for Migrant Construction Workers	A portable, offline-capable health record system that syncs across states so migrant laborers don't lose medical history when they relocate for work.
33	Software	Governance & Public Services	Crowdsourced Pothole and Road Hazard Reporting Platform	A citizen app with photo-geotagging and municipal dashboard integration to prioritize road repairs based on traffic volume and severity.
34	Software	Smart Education	AI Dropout Risk Predictor for Government Schools	A system analyzing attendance, test scores, and socioeconomic indicators to flag at-risk students early and suggest intervention strategies for teachers.
35	Software	Governance & Public Services	Blockchain-Based Land Record Verification for Rural Property Disputes	A tamper-proof digital ledger integrating with existing revenue department records to reduce fraudulent land claims.
36	Software	Governance & Public Services	Multilingual Grievance Redressal Chatbot for Public Schemes	An AI chatbot that helps citizens in regional languages check eligibility and track application status for welfare schemes like PM-Kisan or Ayushman Bharat.
37	Hardware	Agriculture, FoodTech & Rural Development	Low-Cost IoT Soil Moisture and Nutrient Sensor Kit	Solar-powered sensor nodes that transmit real-time soil health data to farmers via SMS, designed for areas with poor connectivity.
38	Hardware	Water & Sanitation	Portable Water Quality Testing Device for Rural Wells	A handheld device using electrochemical sensors to detect contaminants (fluoride, arsenic, bacterial load) with instant readouts, no lab required.
39	Hardware	Agriculture, FoodTech & Rural Development	Solar-Powered Cold Storage Unit for Small-Scale Farmers	A modular, low-maintenance cold chain solution to reduce post-harvest losses of perishable produce in areas without reliable grid power.
40	Hardware	Disaster Management & Climate Resilience	Early Flood Detection Sensor Network for River Basins	Ultrasonic water-level sensors placed along riverbanks that trigger automated alerts to downstream villages before water levels become dangerous.